MONA FAKHRI SALEM

BIOINFORMATICS | COMPUTATIONAL BIOLOGY | BIOSTATISTICS

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PROFILE

Bioinformatics undergraduate at Cairo University with training and academic project experience in molecular docking, sequence analysis, biological databases, transcriptomics, multi-omics, biostatistics, and machine learning on health data. Research interests include computational drug discovery and genomic data analysis.

EDUCATION

Cairo University | B.Sc. candidate — Computer Science (Bioinformatics)

Relevant areas: Computational Biology • Biostatistics • Drug Design • Biological Sequence Analysis

SELECTED RESEARCH & ACADEMIC PROJECTS

- **Molecular Docking of Candidate Compounds Against SARS-CoV-2 Main Protease (6LU7).** Completed as the capstone project for I-Mole Lab bioinformatics training, supported by Cairo University and presented to university faculty. Prepared protein/ligand structures and docked six compounds—cetirizine, N3, ebselen, remdesivir, DMSO, and nitrobenzaldehyde—using AutoDock Vina/AutoDock and Open Babel. Retrieved structures/compound information from PDB, PubChem, and DrugBank; inspected ligand–protein interactions with Discovery Studio Visualizer. Ebselen and remdesivir had the lowest reported docking scores (−6.3); these are computational scores, not experimental evidence of efficacy.
- **Patient Demographic and Health Data Analysis (R).** Analyzed a patient-level dataset using data filtering, missing-value checks, descriptive summaries, and visualizations. Compared disease patterns across age, gender, ancestry, education, and family-size groups, and summarized commuting and daily internet-use distributions.
- **Heart Disease Classification and Model Comparison (Python).** Worked with a 918-record heart-disease dataset and compared Logistic Regression, Decision Tree, Random Forest, and XGBoost classifiers. Used train/validation splits and accuracy to review model performance and tree settings; implemented analysis with pandas, scikit-learn, and visualization libraries.

TRAINING & CERTIFICATES

- **Story of Bioinformatics (Basic Bioinformatics) & Drug Design** — I-Mole Lab, 14 Sep 2024
  Completed both courses and presented the related SARS-CoV-2 molecular-docking project to Cairo University faculty.
- **Introduction to Bioinformatics** — Baheya Foundation, 15 Jul–5 Aug 2024
  Biological databases and data retrieval • sequencing • BLAST • Bash • R-based transcriptome analysis • genomic and multi-omics analysis
- **Introduction to Bioinformatics** — Children's Cancer Hospital Egypt 57357, 13 Aug–20 Sep 2023 (48 hours)
  Biological databases • sequence and multiple-sequence alignment • BLAST • Galaxy • molecular evolution • phylogenetics
- **Python for Genomic Data Science** — Johns Hopkins University via Coursera, 30 Jan 2024
- **Biology Meets Programming: Bioinformatics for Beginners** — University of California San Diego via Coursera, 20 Oct 2023
- **Advanced Learning Algorithms** — Stanford University & DeepLearning.AI via Coursera, 20 Feb 2024

TECHNICAL SKILLS

Bioinformatics methods: Biological databases (PDB, PubChem, DrugBank), BLAST, sequence alignment, transcriptomics, multi-omics, phylogenetics, molecular docking, biostatistics.

Programming & analysis: Python, R, Bash, Biopython, pandas, NumPy, scikit-learn; data cleaning, exploratory analysis, statistical summaries, classification, and visualization.

Scientific software: AutoDock Vina, AutoDock, Open Babel, Discovery Studio Visualizer, Galaxy, Jupyter Notebook.

LANGUAGES

Arabic • English • German